# Progress

## AI-RAN GPU Isolation — Mechanism Deep-Dive

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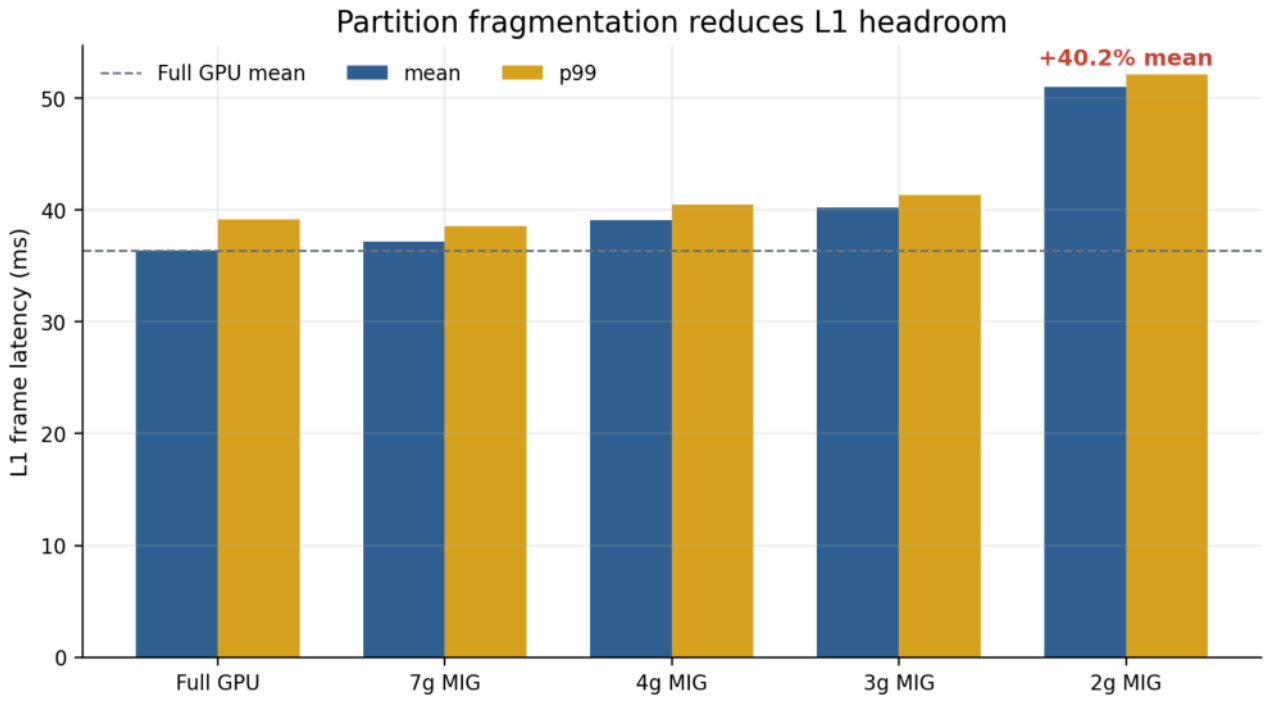
Recap previous results, then drill into the mechanism.

# Recap & Today's Question

From overhead measurement to mechanism identification

- Previous (5/29): MIG single-tenant overhead +13-32ms (Exp1-6)
- Today: AI co-tenant 환경에서 비용은 어디서, 왜 발생하는가?
- Scale: n=500-1000/condition (vs n=20 last time) — statistical power 25×
- Platforms: CloudLab d8545 driver 550 + Perlmutter cross-validation
- Goal: identify the hardware resource behind the overhead

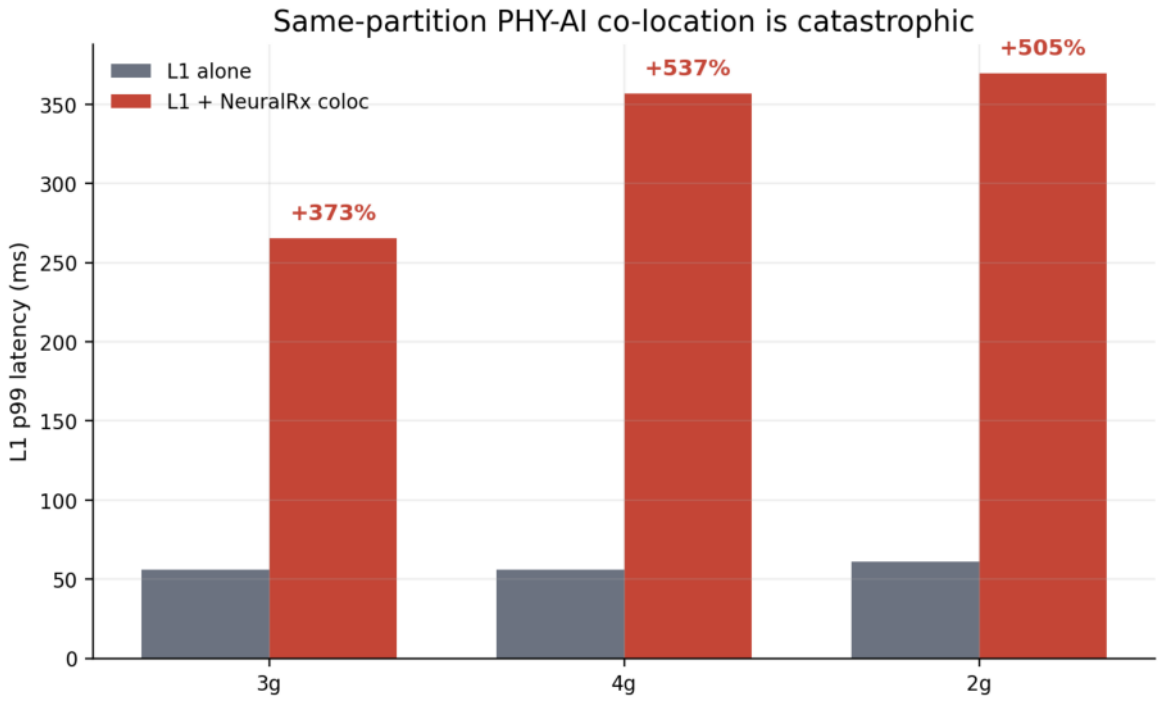



Cross-partition은 봤는데, same-partition은 어떻게 되나?

# Exp7) Same-Partition Coloc = Catastrophic Collapse

Cross-partition stable, same-partition collapses 8×

- Cross-partition stays ~45ms, same-partition coloc collapses to ~357ms (8×)
- 12+ workloads (chanpred, NeuralRx, qwen, xapp, ResNet) all collapse to ~357ms
- Workload-invariant → systemic mechanism, not a workload-specific trick
- n=500/condition, std=9ms (stable, reproducible)
- First evidence: cost is from placement, not the AI's nature



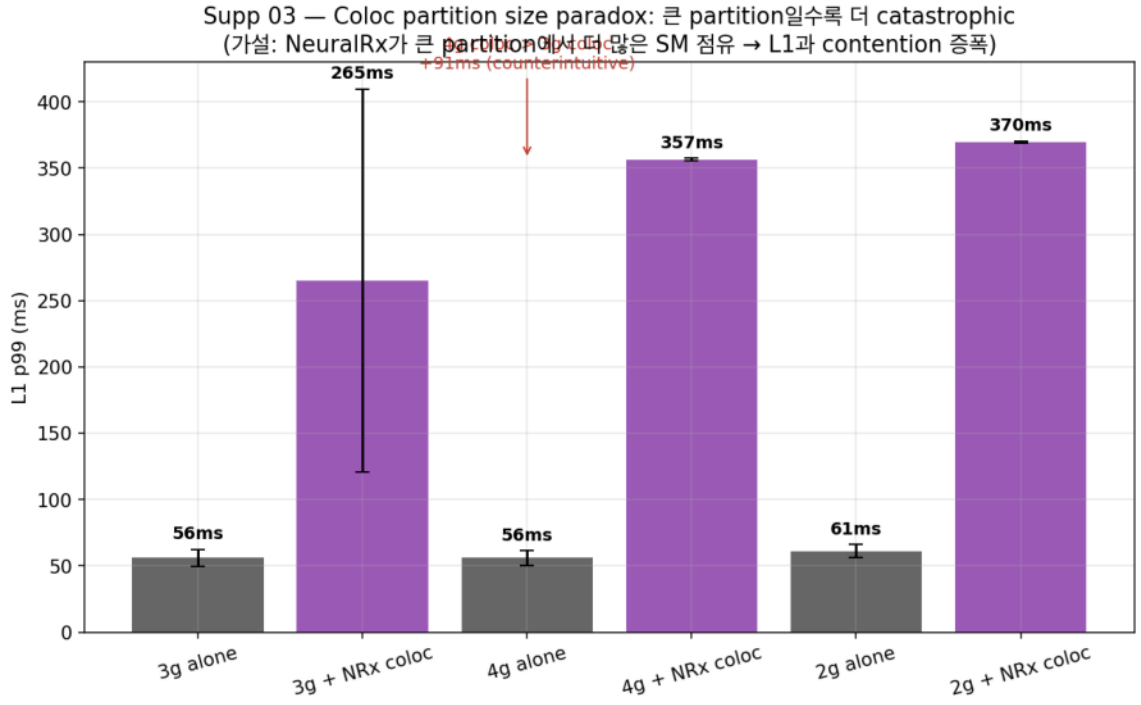

| Workload | Cross-part | Coloc |
|---|---|---|
| chanpred | 45 | 357 |
| NeuralRx | — | 360 |
| qwen | — | 357 |
| xapp | — | 358 |
| ResNet | — | 358 |

큰 partition으로 SM/HBM을 더 주면 해결되나?

# Exp8) Larger Partition = MORE Catastrophic

Counter-intuitive partition size paradox

- Larger slice does NOT help; 4g coloc (371ms) WORSE than 2g (369ms)
- Resource bandwidth share is decisive — but NOT via partition allocation
- Rejects the obvious 'give more SMs' mitigation hypothesis
- Bottleneck is outside the partition boundary
- First hint: the costly resource is chip-global, not slice-local



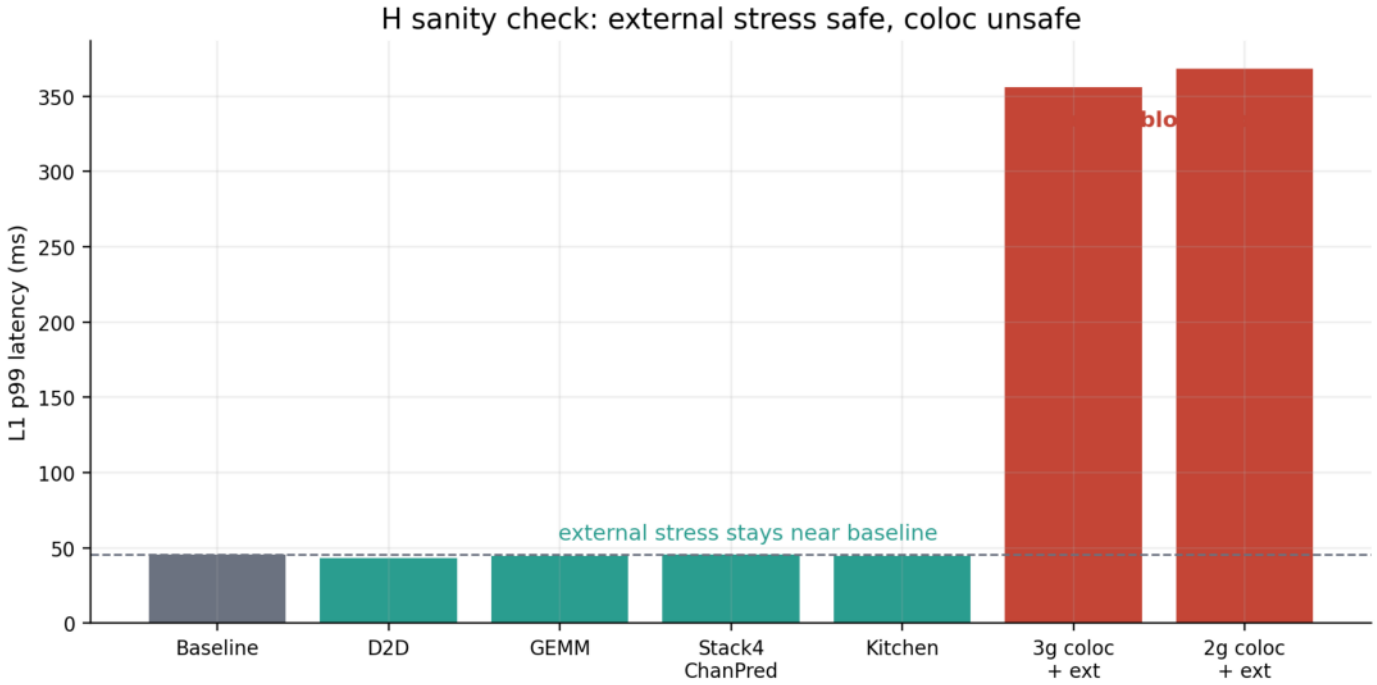

| Partition | Coloc latency (ms) |
|---|---|
| 2g.10gb | 369 |
| 3g.20gb | 361 |
| 4g.20gb | 371 |

Placement는 binary인가, gradient인가?

# Exp9) Placement = Binary Decision, No Soft Transition

One config bit decides 8× collapse

- Same workload (chanpred), only placement differs: cross 44ms ↔ coloc 357ms
- No middle ground, no gradient between the two regimes
- One config mistake = 8× collapse — operational fragility
- Implication: there exists a single shared resource flipping state
- Now we must find WHICH resource — switch to NSYS profiling



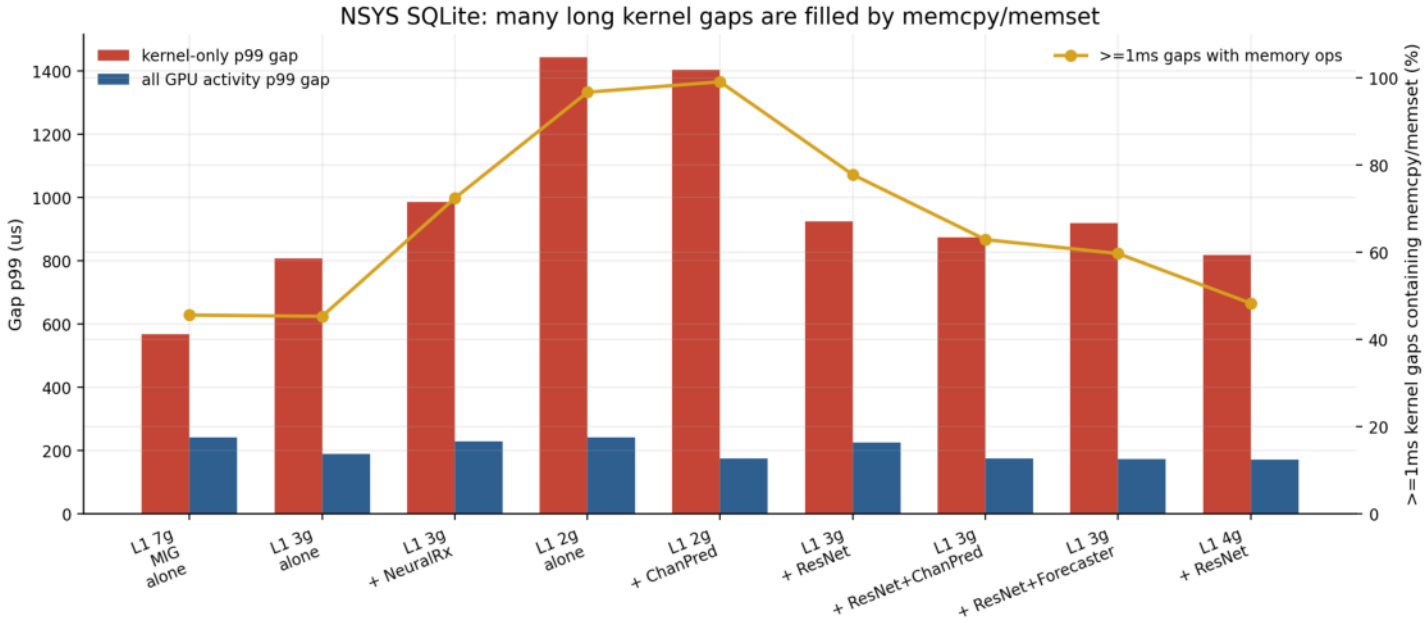

| Condition | Placement | Latency (ms) |
|---|---|---|
| H_F1 | cross-partition | 44 |
| H_G1 | same-partition coloc | 357 |

같은 hardware에서 왜 8× 차이? 무엇이 변하는가?

# NSYS Layer 1 — Gap is at Memcpy/Memset Boundary

Cost is NOT inside the kernel — it is between kernels

- L1 kernel duration is unchanged under coloc
- Inter-kernel 'gap' grows by exactly the missing latency
- Gap is NOT random idle — concentrated at memcpy/memset boundaries
- Aggregated across 30 conditions → robust, not cherry-picked
- The bottleneck is a memory-transfer wait, not compute

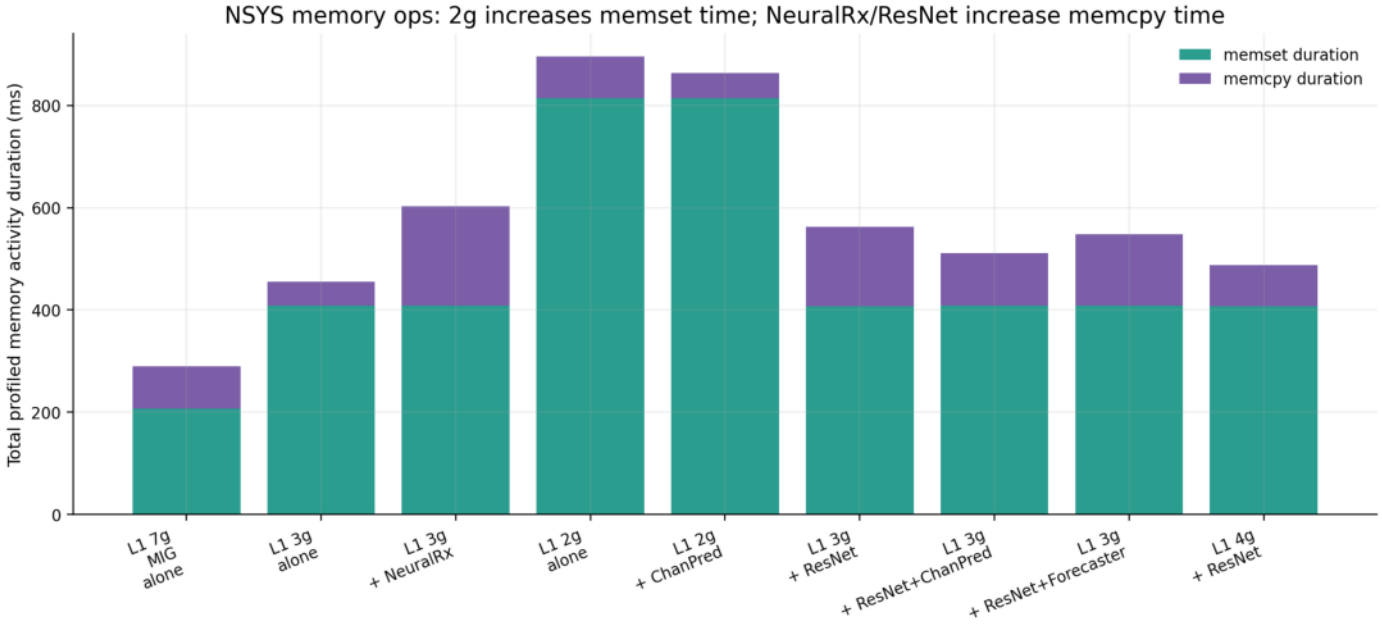



Gap이 늘었다면 — op count가 늘었나, op duration이 늘었나?

# NSYS Layer 2 — Same Ops, Just Slower

Rejects 'more work' hypothesis — count invariant, duration inflates

- memcpy call count: 5,778 → 5,778 (0% change)
- memset count: unchanged
- memcpy avg duration: 4.2 → 9.8µs (+133%)
- AI is NOT giving L1 more work — same work, blocked longer
- Confirms: cost is waiting, not extra activity



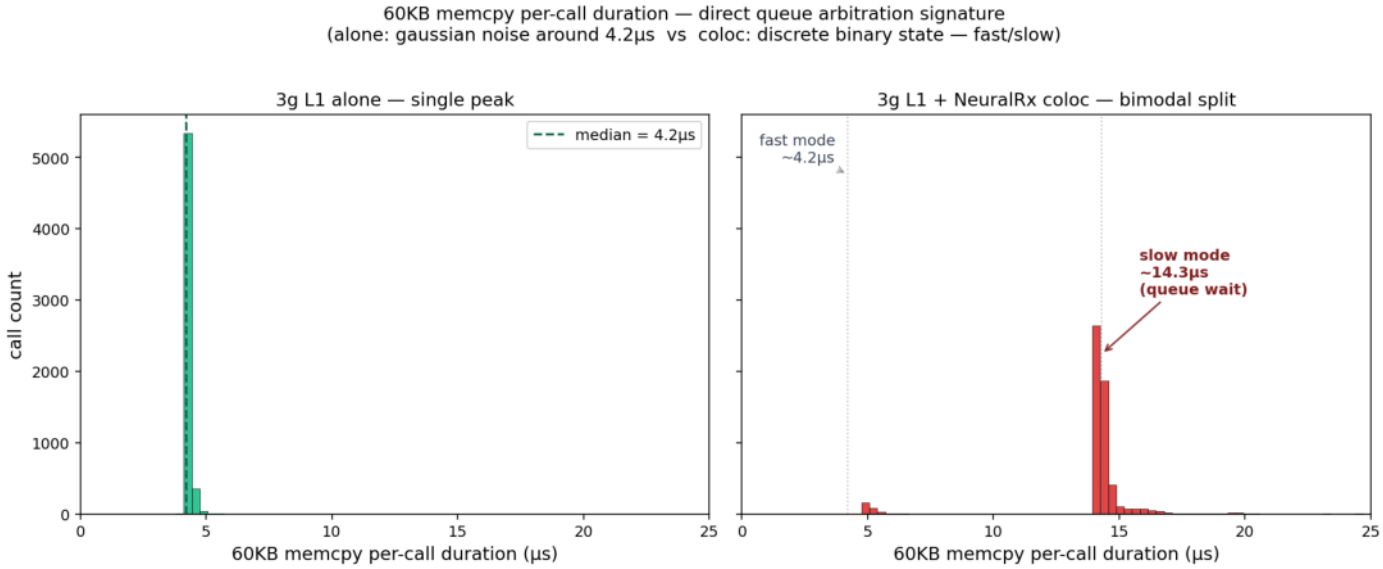

| Metric | Alone | Coloc | Change |
|---|---|---|---|
| memcpy count | 5,778 | 5,778 | 0% |
| memcpy avg µs | 4.2 | 9.8 | +133% |

왜 같은 op가 더 오래 걸리나? 무엇이 op를 막는가?

# NSYS Layer 3 — Bimodal Signature = Direct Queue Evidence

Per-call duration splits into two discrete modes

- Per-call 60KB memcpy duration splits into TWO discrete modes
- Alone: single gaussian peak at 4.2µs
- Coloc: bimodal 4.2µs ↔ 14.3µs (fast=queue empty, slow=queue waiting)
- Discrete state, not gaussian noise → arbitration signature
- This is the smoking gun for a shared serialized queue



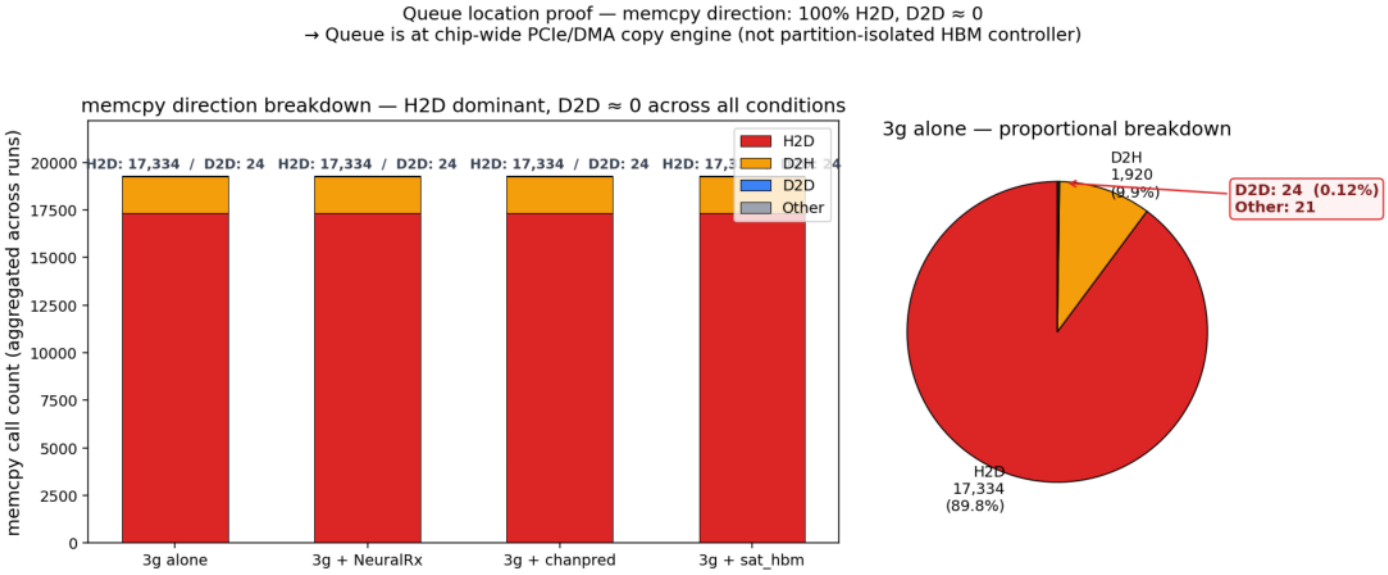

| Mode | Alone (µs) | Coloc (µs) |
|---|---|---|
| Fast peak | 4.2 | 4.2 |
| Slow peak | — | 14.3 |
| Distribution | unimodal | bimodal |

어떤 queue? 위치를 알아야 격리 가능성을 판단한다.

# NSYS Layer 4 — Chip-Wide PCIe/DMA Copy Engine

CLIMAX — the shared resource is identified

- Direction breakdown: H2D 17,334 (89.8%) / D2H 1,920 (9.9%) / D2D 24 (0.12%)
- D2D ≈ 0 → queue is NOT at HBM controller (HBM carries D2D)
- 100% host-path → queue is at chip-wide PCIe/DMA copy engine
- This single resource is shared across the WHOLE chip — partition cannot isolate it
- Explains: Slide 3 paradox, Slide 4 binary, Slide 7 bimodal — all consistent



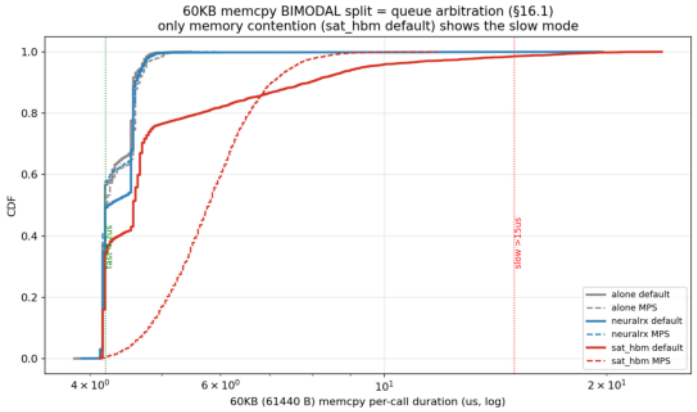

| Direction | Count | Share |
|---|---|---|
| H2D | 17,334 | 89.8% |
| D2H | 1,920 | 9.9% |
| D2D | 24 | 0.12% |
| Other | 21 | 0.11% |

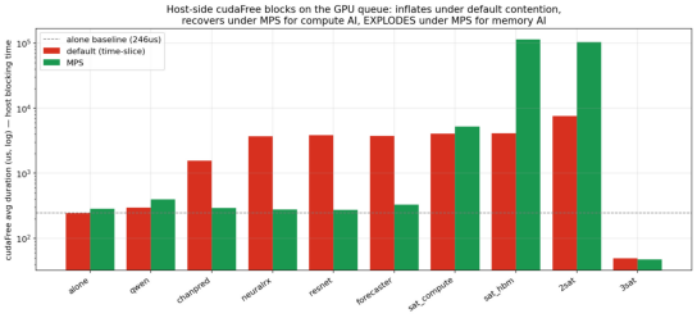
이건 MIG 결함인가, 아니면 NVIDIA stack 전체의 문제인가?

# Cross-Platform Validation — Mechanism is Universal

Same signature on Perlmutter (no-MIG, different driver, different cluster)

- (A) Bimodal signature reproduced: 4.2 ↔ 16.8µs on Perlmutter
- (B) cudaFree blow-up: alone 246µs → NeuralRx 3,752µs (15×) → sat_hbm MPS 115,506µs (115ms!)
- (C) correlationId causal proof: 60-82% of GPU gap overlaps EXACTLY with cudaFree host-blocking
- Conclusion: PCIe/DMA queue is an NVIDIA-stack-wide fundamental property
- Not a MIG bug — disabling MIG does not escape it



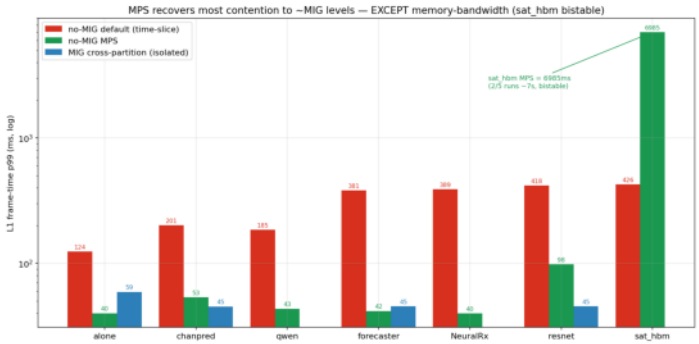

| Metric | Alone | NeuralRx default | sat_hbm MPS |
|---|---|---|---|
| cudaFree avg | 246µs | 3,752µs (15×) | 115,506µs (115ms) |
| Bimodal peaks | 4.2µs | 4.2 ↔ 16.8µs | — |
| Gap↔cudaFree overlap | — | 60-82% | — |

그럼 MPS는 cudaFree wait을 우회할 수 있나?

# MPS — Recovers Compute AI, Catastrophic for Memory AI

Same trap, different trigger

- Compute AI: MPS RECOVERS — NeuralRx 389→40ms, forecaster 381→42ms, qwen 185→43ms
- Memory AI: MPS CATASTROPHIC — sat_hbm 426 → 6,985ms (bistable 7-sec freeze)
- Why recover? True concurrent exec → compute parallelizes → cudaFree returns fast
- Why collapse? Memory hog starves DRAM → device truly stuck → cudaFree blocks 115ms → 7s
- No universal escape — same mechanism, different trigger





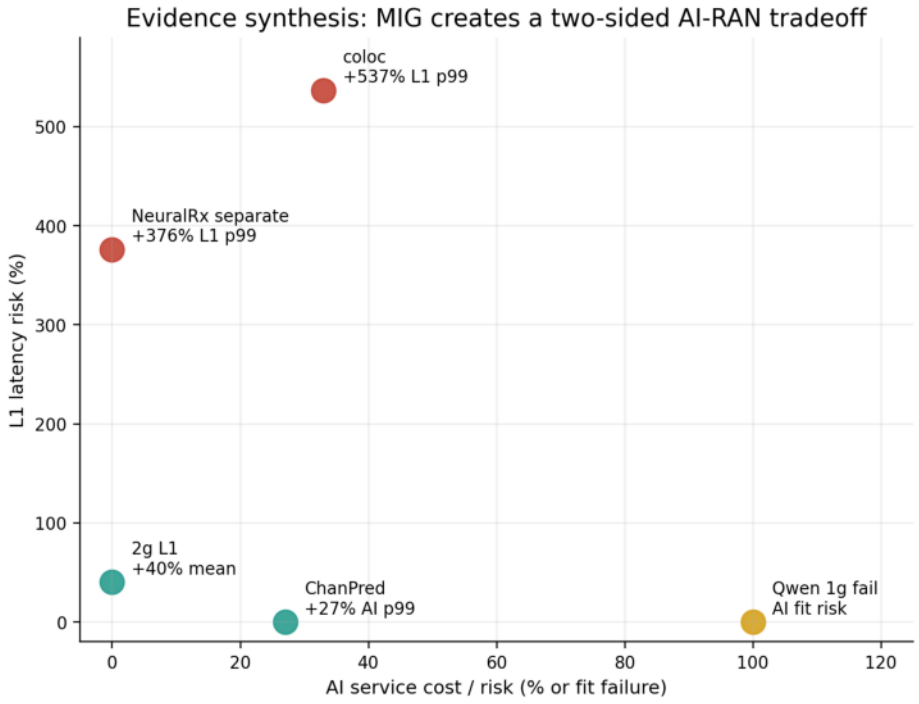
| Workload | Default | MPS | MIG cross |
|---|---|---|---|
| NeuralRx | 389ms | 40ms | 197ms |
| forecaster | 381ms | 42ms | — |
| qwen | 185ms | 43ms | — |
| sat_hbm | 426ms | 6,985ms | — |

이 모든 발견을 어떻게 하나의 framework로 종합하나?

# Cost Decomposition — 3-Platform Unified Framework

Two hardware-level costs, not one — the paper contribution

- MIG overhead = TWO hardware-level costs, not one monolithic 'overhead'
- Structural (memset) = partition HBM bandwidth share — controllable by partition choice
- Contention (memcpy) = chip-wide PCIe/DMA queue — NOT controllable
- 3 platforms (MIG / no-MIG / MPS) each avoid one cost but fail at the other
- No universal isolation exists in current NVIDIA stack




| Cost type | Hardware | MIG isolates? |
|---|---|---|
| Structural | HBM bandwidth | Yes |
| Contention | PCIe/DMA copy engine | NO |

이 framework의 운영 함의(design rules)는?

# Design Rules & Next Steps

Measurement-validated operational guidance

- Rule 1: L1과 AI는 반드시 다른 MIG partition (single mistake → 8× collapse)
- Rule 2: Partition 크기는 L1 SM/HBM headroom으로만 결정 (coloc 회피용 무의미)
- Rule 3: AI workload classification 필요 (memory-bound AI는 MPS에도 위험)
- Rule 4: MIG isolation ≠ chip-wide resource isolation (paper main contribution)
- Next: CloudLab Phase 4 (MIG-off + MPS cross-check), N-AI scaling law, driver 550 baseline

| Next Step | Target |
|---|---|
| CloudLab Phase 4 | MIG off + MPS — cross-check Perlmutter on same HW |
| N-AI scaling law | L1 + N AI same-partition, direct measurement |
| Driver 550 baseline | Reproduce on CloudLab d8545 |

End.

# Thank You

Questions & Discussion

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